

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without permission of BIS or used as an Indian Standard)

भारतीय मानक मसौदा
सिल्वरमैन प्रकार की यकृत ऊतक नमूना लेने के लिए सुई - विशिष्टि
[IS 7387 का पहला पुनरीक्षण]

Draft Indian Standard
Silverman Type Liver Biopsy Needle - Specification
[First Revision of IS 7387]

ICS 11.040.25

Hospital Equipment and Surgical Disposable
Products Sectional Committee, MHD 12

Last date for comments:
12 September 2026

FOREWORD

(Formal clause will be added later)

This standard was originally published in 1974. The first revision of this standard has been brought out to align it with recent developments in the material requirements, designation of materials, removal of tests for sharpness of biopsy needle point, addition of tolerance to dimensions, sampling method and to bring the standard in line with the latest style and format of referred standards.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This standard specifies the dimensional and other requirements for Silverman type liver biopsy needle.

2 REFERENCES

The standards given below contain provisions, which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No.</i>	<i>Title</i>
IS 6911: 2026	Stainless Steel Plate, Sheet and Strip - Specification (<i>third revision</i>)
IS 17964 (Part 7): 2025	Small-bore connectors for liquids and gases in healthcare applications: Part 7 Connectors for intravascular or hypodermic applications
IS 1068 : 2025/ ISO 1456 : 2009	Metallic and other inorganic coatings — Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium (<i>fourth revision</i>)
IS 4905: 2015 ISO 24153: 2009	Random sampling and randomization procedures (<i>first revision</i>)

3 TERMINOLOGY

3.1 Lug - The small projection on the Luer connector of specialized needles to indicate the bevel's orientation.

3.2 Luer Connector - The tapered fitting that creates a leak-free, secure and detachable connection.

3.3 Stilette - It is a thin, solid rod that is inserted into a hollow medical needle (cannula).

3.4 Adopter - The adopter is a device used to securely connect a needle to another component, such as a syringe, IV tubing or blood collection tube holder.

3.5 Cannula - The cannula is a tube that can be inserted into the body, often for the delivery or removal of fluid or for the gathering of samples.

4 MATERIAL REQUIREMENTS

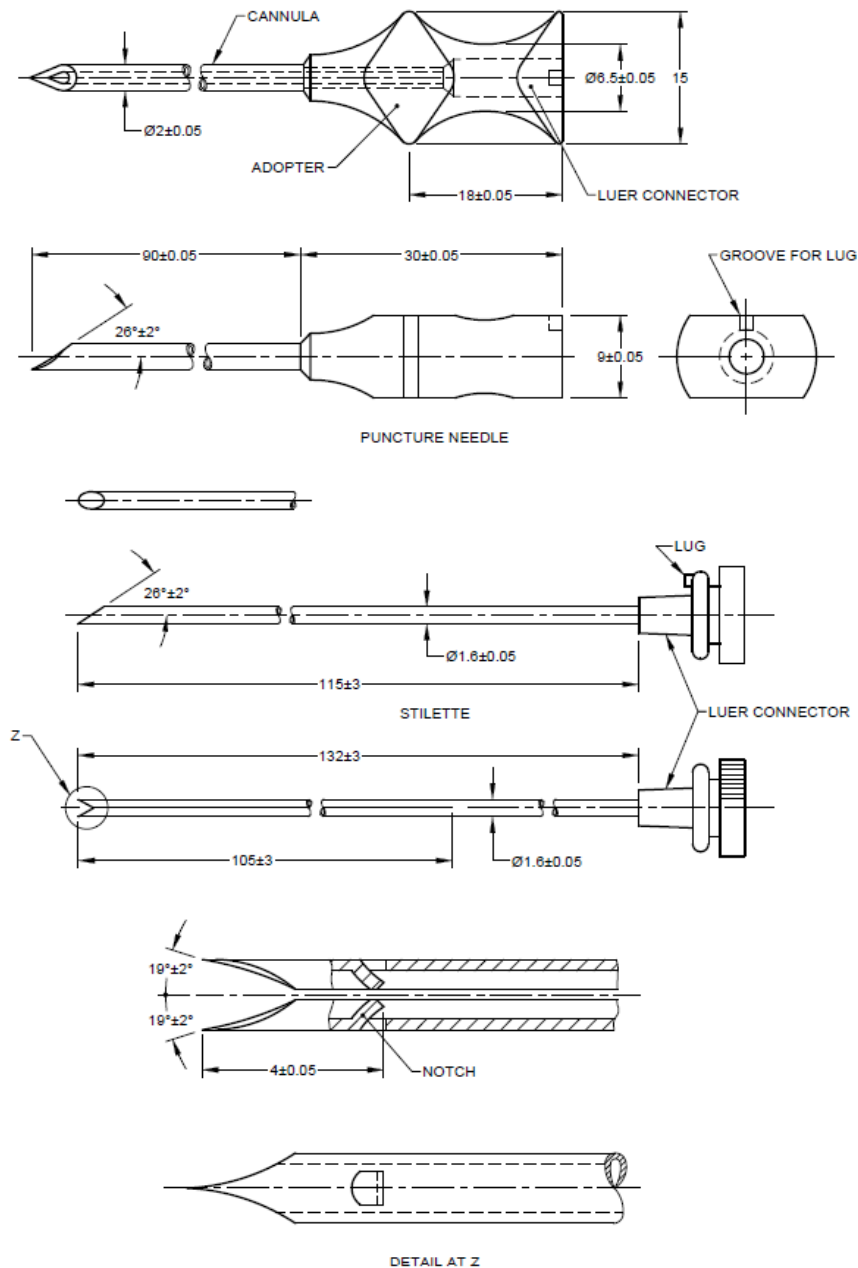
4.1 The cannula shall be made from stainless steel drawn tubing which conform to designation X07Cr18Ni9S or X04Cr17Ni12Mo2 of IS 6911.

4.2 The Luer Connector materials requirements shall be as per Annex G of IS 17964 (Part 7).

4.3 The stilette wire shall be made from stainless steel wire.

5 SHAPE AND DIMENSIONS

The shape and dimensions shall be as shown in Fig. 1.



All dimensions in millimetres

Fig. 1 LIVER BIOPSY NEEDLE

6 CONSTRUCTION REQUIREMENTS

6.1 The conical fitting shall be of the Luer type connector and shall be in accordance with IS 17964 (Part 7).

6.2 The cannula shall be pushed well into the cavity of the Luer connector but not extending into the conical portion and securely swaged. The cannula and the Luer connector shall be concentric and well aligned.

6.3 The puncture and biopsy needle shall be sharpened on three aspects as shown in Fig. 1. The points on both the needles shall be well defined and free from any feather edge, irregularity and burrs when examined under 10X to 15X magnification.

7 WORKMANSHIP AND FINISH

7.1 Workmanship

7.1.1 The cannula shall be smooth both outside and inside and free from pits, cracks, tool marks, burrs, embedded foreign matter and other surface defects.

7.1.2 The stilette wire shall be smooth, bright and free from foreign matter and shall slide smoothly into the cannula.

7.2 Finish

The Luer connector shall be free from pits, burrs, cracks, sharp edges, tool marks, roughness and other surface defects. It shall be plated both inside and outside chromium over nickel conforming to 5.6 of IS 1068 and polished. The plating shall be such as to withstand the corrosion test given at 6.5 of IS 1068.

8 TESTING REQUIREMENTS

8.1 Leakage Test for Puncture Needle

The needle shall be fit to a tested syringe and connect the syringe to a source of water on which pressure could be applied. The water through the needle point shall be started and sealed by piercing into a piece of solid rubber and applying a pressure of 3 Bar for 10 seconds. No leakage shall be observed at the joint of the Luer connector and the syringe or the cannula and the Luer connector.

8.2 Sharpness of Puncture Needle Point

The aluminium foil of 0.02 ± 0.01 mm thickness is cleaned with alcohol or benzene and mounted suitably so that it is spread taut and the needle may penetrate vertically. The needle shall be kept vertically with its point touching the foil and ensuring that the weight of the needle did not act on the foil. Then the load on the Luer connector shall be gently started with one gram. The load on foil shall be increased gradually till 30 seconds. The needle shall penetrate the aluminium foil with a test load less than 30 ± 0.5 g.

8.3 Flexibility of Biopsy Needle

The split portions of the biopsy needle shall be pulled so that the tips separate by 30 mm and released. This shall be done 5 times. The needle shall not break or take a permanent set.

8.4 Security of Swaging

The swaging of the cannula of the puncture needle with the Luer connector shall be tested by applying a pull of 9 kg for one minute. The cannula shall not come out of the Luer connector and it shall not become loose.

8.5 Corrosion Resistance

The complete needle shall be immersed in a 10 percent solution of citric acid at room temperature for 5 hours. It shall then be boiled in distilled water for 30 minutes and cooled while immersed in the same for 48 hours. The cannula or Luer connector shall show no corrosion. The test shall be conducted in a glass container.

8.6 Freedom from Foreign Matter

The cannula and the Luer connector shall be examined under good illumination with a magnifying lens, they shall be free from foreign matter, metal particles or chips, pits, burrs and other defects.

9 SAMPLING PROCEDURE

The Sampling procedure and criteria for conformity for liver biopsy needle may be as agreed to between the purchaser and the supplier. A recommended sampling plan is as given below:

9.1 Lot

The lot includes consignment with all liver biopsy needles of the same design and size and manufactured under similar conditions of production shall be grouped together to constitute a lot.

9.2 Selection of Sample

9.2.1 The number of liver biopsy needles to be selected from the lot shall depend upon the size of the lot and shall be in accordance with col 1 and col 2 of Table 1.

9.2.2 The needles shall be selected from a lot at random. The procedure to ensure the random selection of the unit shall be in accordance with IS 4905.

9.3 Number of Tests and Criteria for Conformity

The needles selected at random according to col 1 and col 2 of Table 1 shall be subjected to the tests for dimensional, construction and testing requirements. Any needle failing to satisfy any of the tests, shall be considered as defective. The lot shall be considered as conforming to the requirements if the number of defectives found in sample is less than or equal to the acceptance number (see col 4 of Table 1) and shall be rejected if it is greater than or equal to the first rejection number (see col 5 of Table 1). If the number of defectives lie between the acceptance and first rejection number, the second sample of the same size (see col 3 of Table 1) shall be chosen at random and tested. If the number of defectives found in the combined samples is greater than or equal to the second rejection number (see col 6 of Table 1), the lot shall be rejected, otherwise the lot shall be accepted.

Table 1 SAMPLE SIZE AND CRITERION FOR CONFORMITY

Lot Size Rejection	First Sample	Second Sample	Acceptance Number	First Rejection Number	Second Number
(1)	(2)	(3)	(4)	(5)	(6)
Up to 50	5	5	0	2	2
51-100	8	8	0	2	2
101-300	13	13	0	2	2
301-500	20	20	0	2	3
501 and above	32	32	0	3	4

10 MARKING

10.1 The instrument container shall be legibly and indelibly marked with the manufacturer's name, initials or trade-mark and the country of manufacture and also marked with word 'STERILE'. In case of stainless steel, the word 'SS' shall also be marked on liver biopsy needle.

10.2 The Luer connector of the liver biopsy needle shall be marked with the diameter of the needle and the manufacturer's identification mark.

10.3 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed there under, and the product(s) may be marked with the Standard Mark.

11 PACKAGING

The liver biopsy needle shall be packed as agreed to between the purchaser and the manufacturer. Each individual package shall contain only one needle and may be marked with the name of the item, and identification of the source of manufacture and batch number and any other marking if so desired.